

Q. An IP Packet has arrived with the first 8 bits as shown 01000010. The receiver discards the packet. Why?

Solution:

Given 8 bits are:

0100 0010

Ver HLEN

HLEN=0010

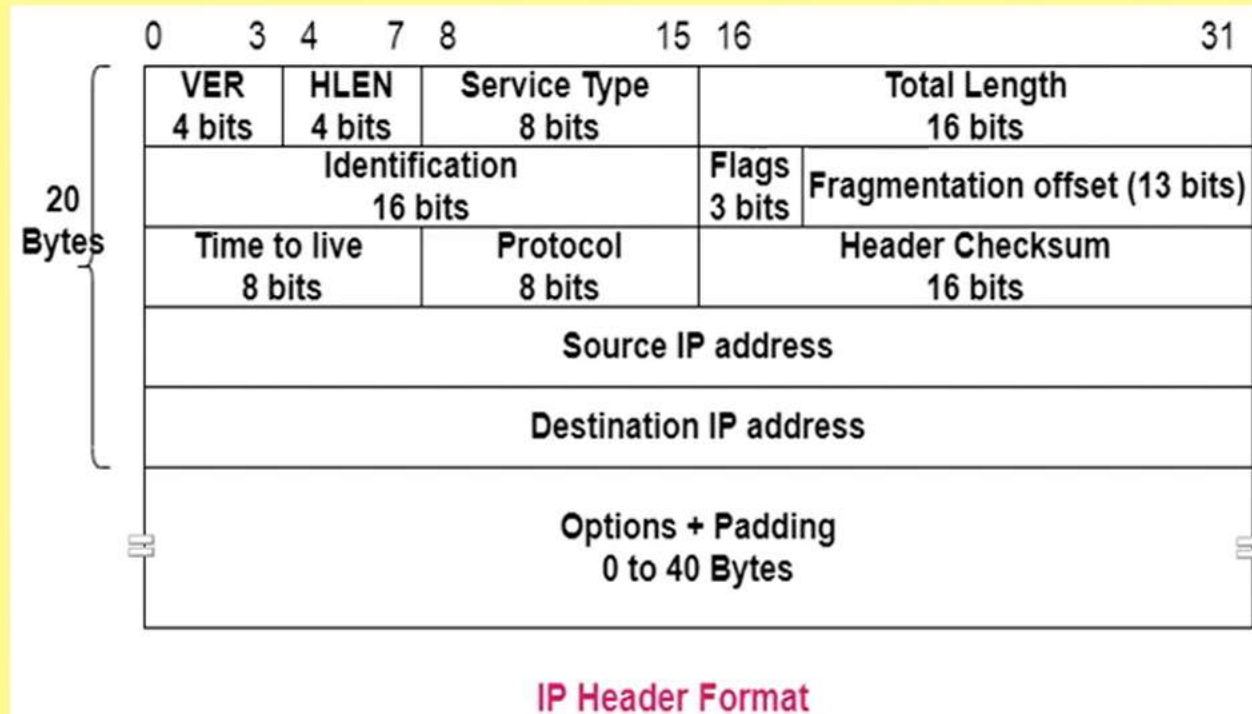
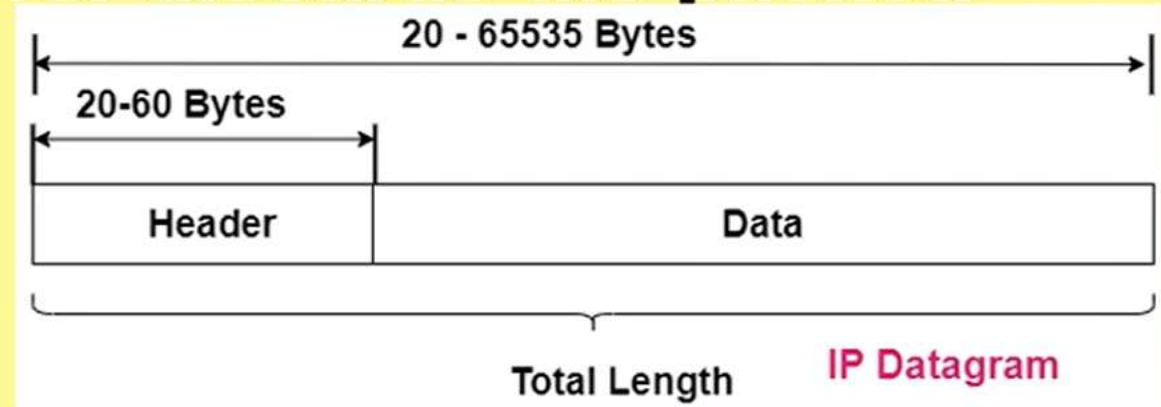
= 2 Hex

= 2 Dec.

Header length (size) = HLEN x 4 bytes

Therefore header length in bytes = 2 x 4 bytes = 8 bytes (Invalid)

Minimum size of IP header should be 20 bytes



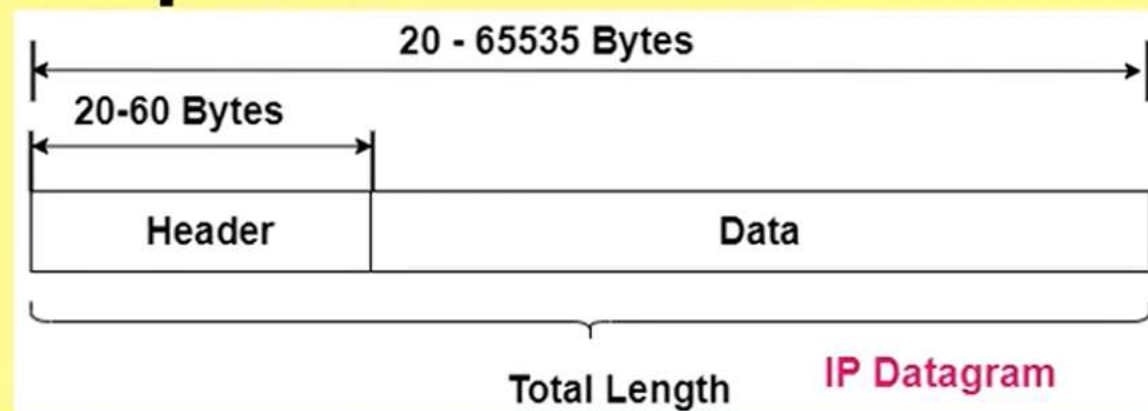
Q.2 In an IP packet the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

Solution:

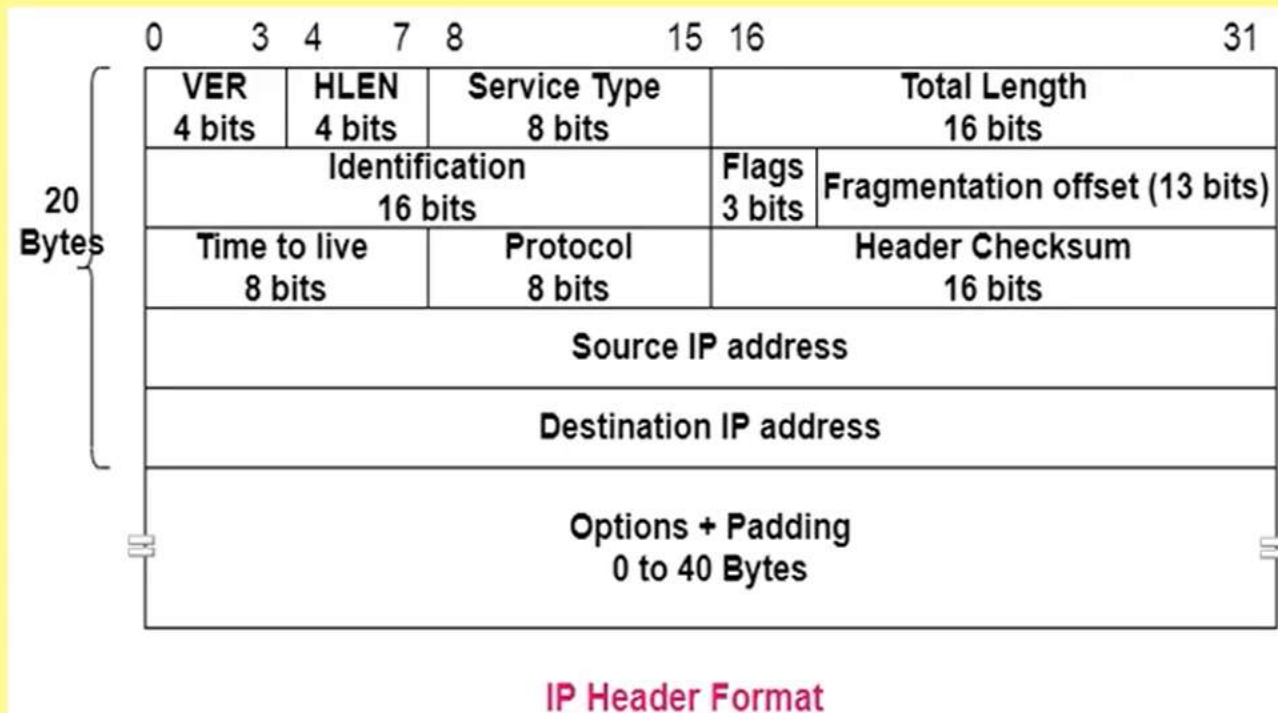
Header length (size) = HLEN x 4 bytes

Options = actual header size – fixed header size

= 32 bytes – 20 bytes = **12 bytes**



HLEN in binary	HLEN in Hex.	HLEN in Dec.	Header length (size) in bytes	Number of bytes in options
1000	8	8	8 x 4 bytes = 32 bytes	32 bytes – 20 bytes (fixed header) = <u>12 bytes of options.</u>



Q. In an IP packet the value of HLEN is 5_{16} and the value of the total length field is 0028_{16} . How many bytes of data are being carried by this packet?

Solution:

$$\text{HLEN} = 5_{16} = 5 \text{ D}$$

Header size (length) = HLEN X 4 bytes

Therefore header length

$$= 5 \times 4 \text{ bytes}$$

$$= 20 \text{ bytes}$$

$$\text{Total length} = 0028_{16}$$

$$= 40 \text{ D}$$

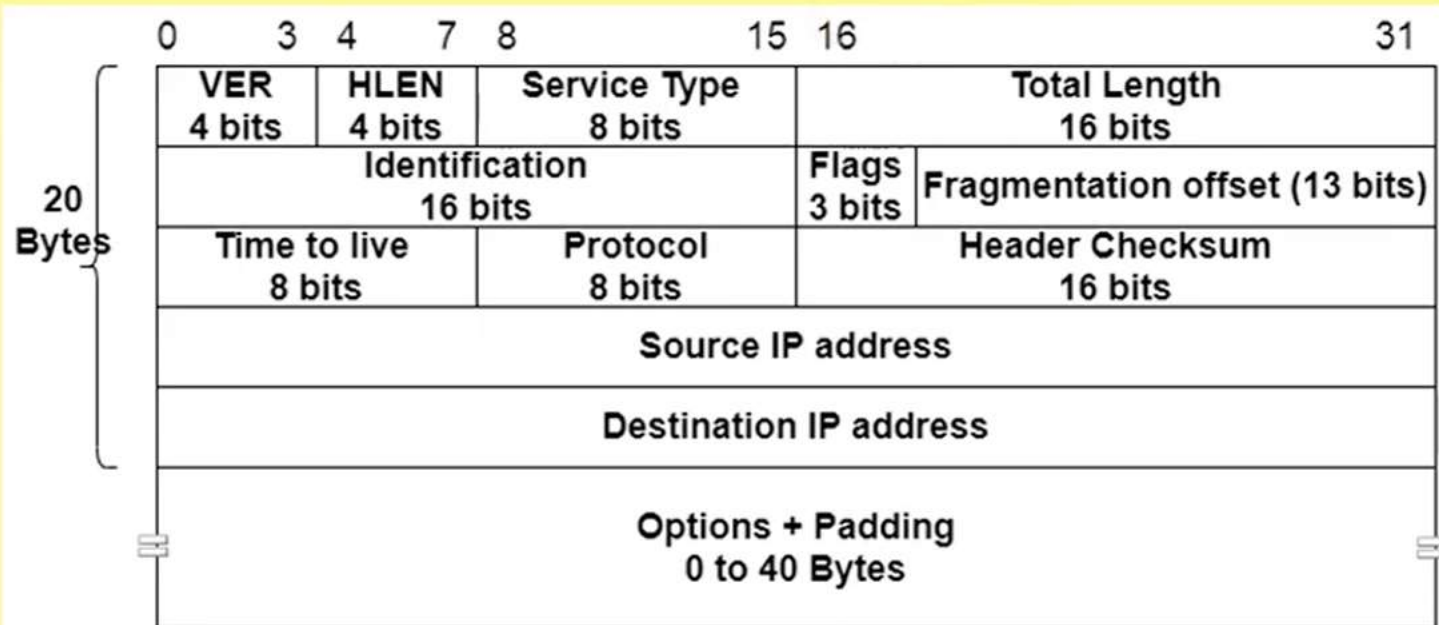
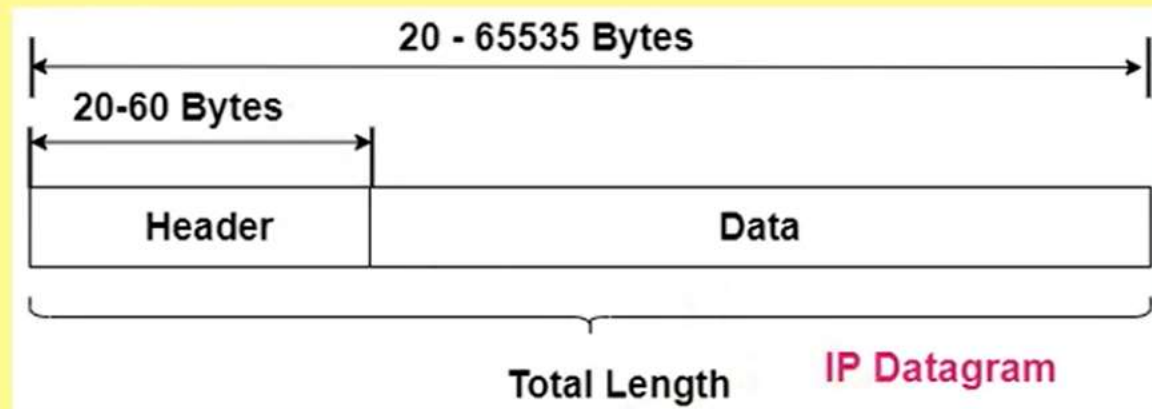
$$= 40 \text{ bytes}$$

Therefore,

$$\text{Data} = \text{total length} - \text{header length}$$

$$= 40 \text{ bytes} - 20 \text{ bytes}$$

$$= 20 \text{ bytes}$$



IP Header Format

Q. An IP packet has arrived with the first few hex digits as : 45000028000100000102..... How many hops can this packet travel before being dropped? The data belongs to which upper layer protocol?

Solution:

4 5 00 0028 0001 0000 01 02

Ver = 4, HLEN = 5, ST = 00,

TL = 0028, Identi. = 0001,

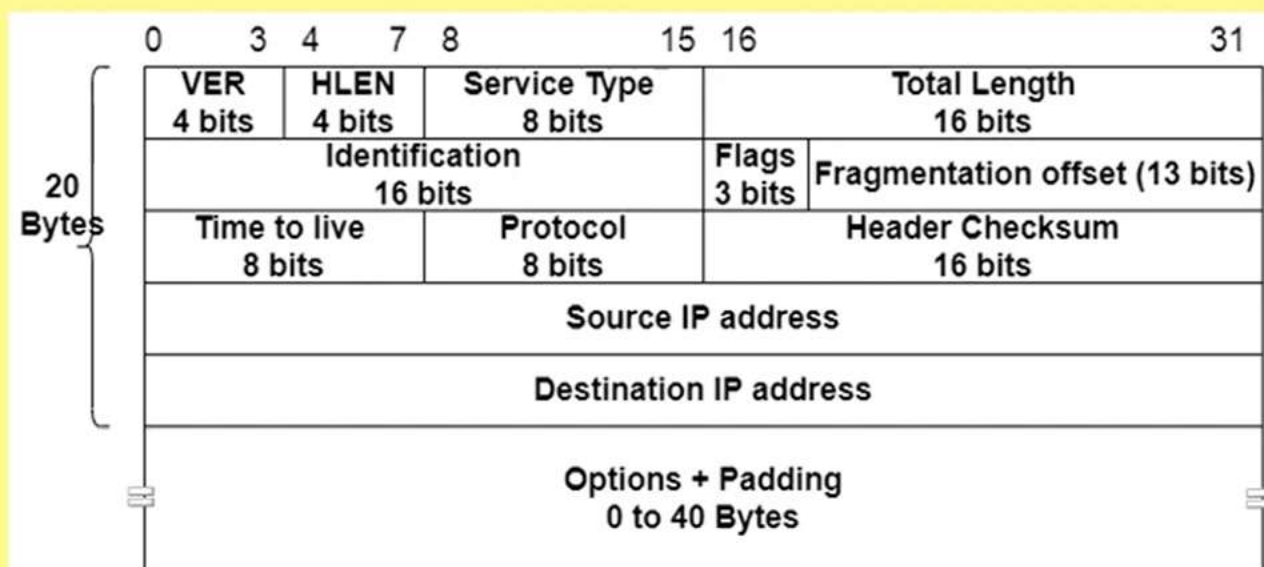
Flags and frag. Offset = 0000

TTL = 01, Protocol = 02

The packet can travel to one router

At the max because TTL field will become zero at this router. So, the packet will not be Forwarded further.

The data belongs to IGMP protocol.



IP Header Format

Value	Protocol
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

Q. Calculate the HLEN value if the total length is 1200 bytes, 1176 of which is data from upper layer.

Solution:

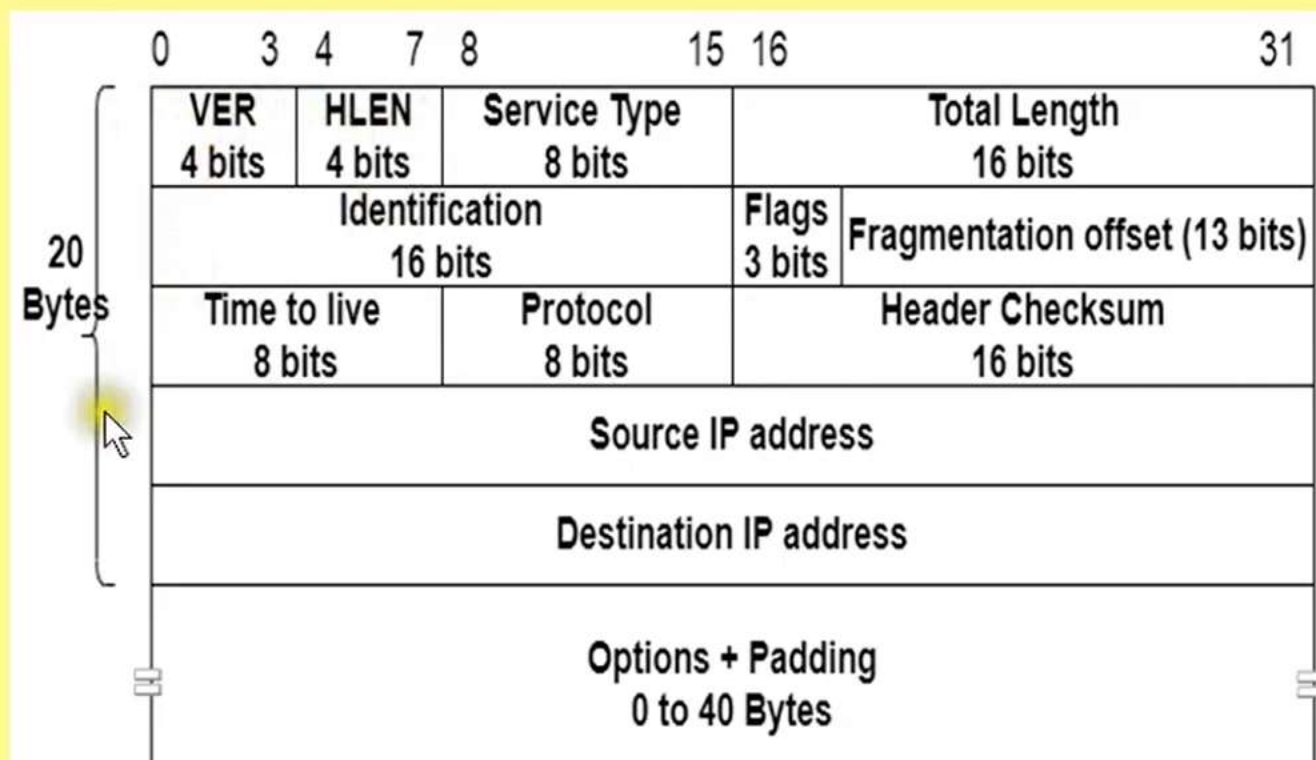
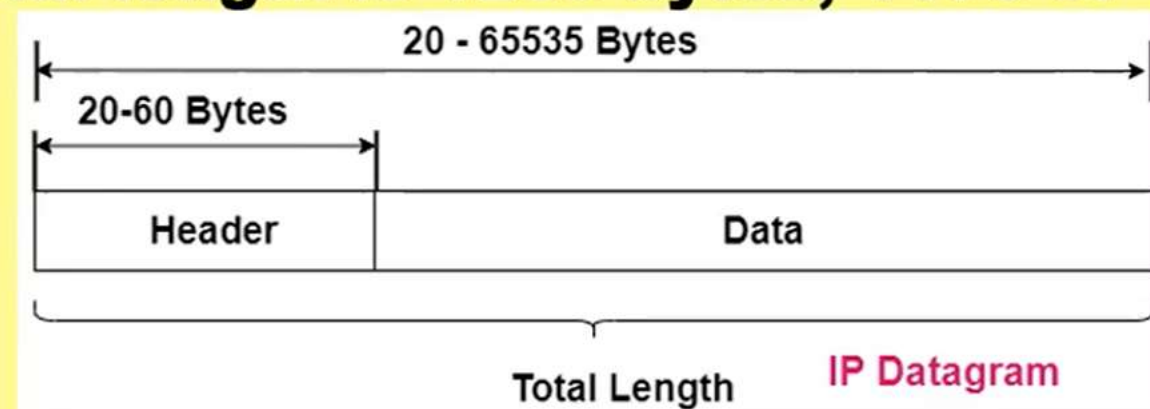
Total length = header length + data length

Therefore, header length = total length - data length
= 1200 bytes - 1176 bytes
= 24 bytes

Now, Header length = HLEN x 4 bytes
Therefore, HLEN = header length / 4 bytes
= 24 bytes / 4 bytes
= 6 D

Therefore HLEN = 0110 in binary

Fixed header size = 20 bytes
Therefore options = 4 bytes

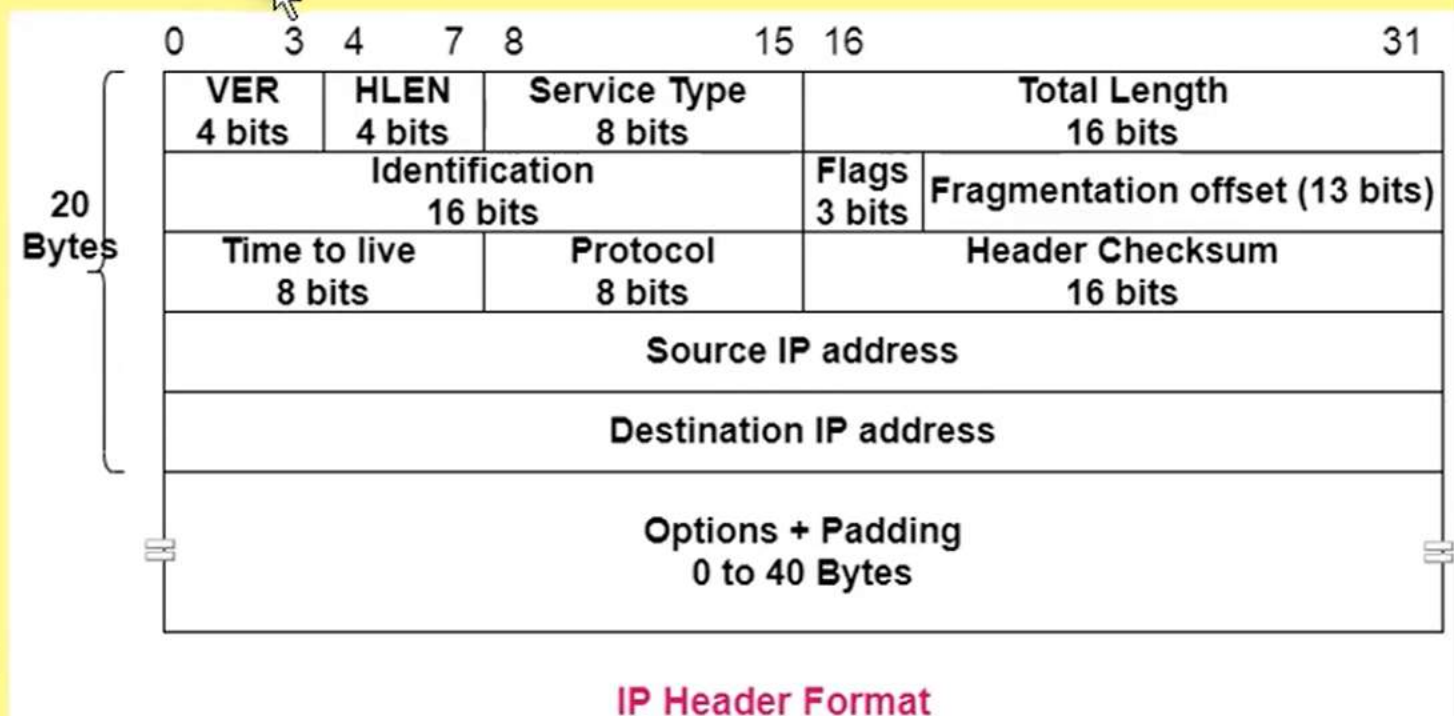


Q. What is the maximum number of routers that can be recorded if the timestamp option has a flag value of 1? Why?

Solution:

TTL = 1

This implies that the maximum Number of routers that can be recorded is 1 as the packet can travel to **maximum one router at which the TTL value will become 0 And the packet will not be forwarded Further.**



Q. The value of HLEN in an IP datagram is 7. How many option bytes are present?

Solution:

HLEN = 7

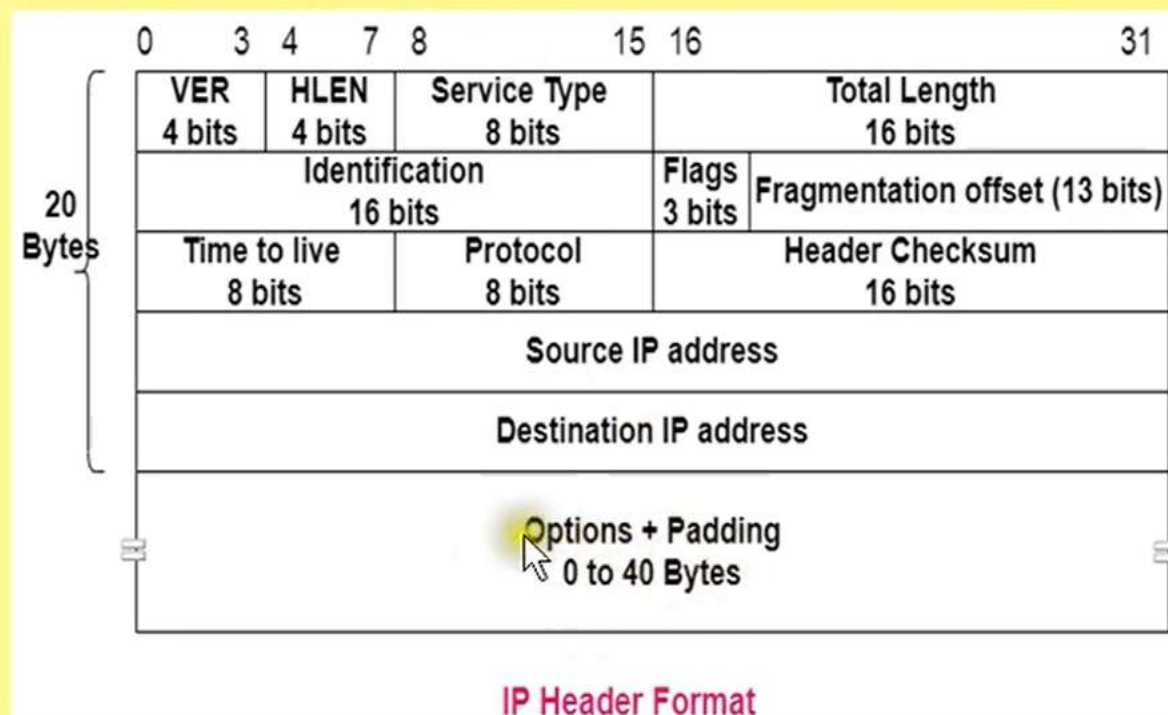
Therefore,

Header length (size) = 7 x 4 bytes = 28 bytes

Fixed IPv4 header size is 20 bytes

Therefore,

**Options = header length – fixed header size
= 28 bytes – 20 bytes
= 8 bytes**



Answer:
Options = 8 bytes

Q. The size of the option field of an IP datagram is 20 bytes. What is the value of HLEN? What is the value in binary?

Solution:

Options = 20 bytes

Fixed IP header size = 20 bytes

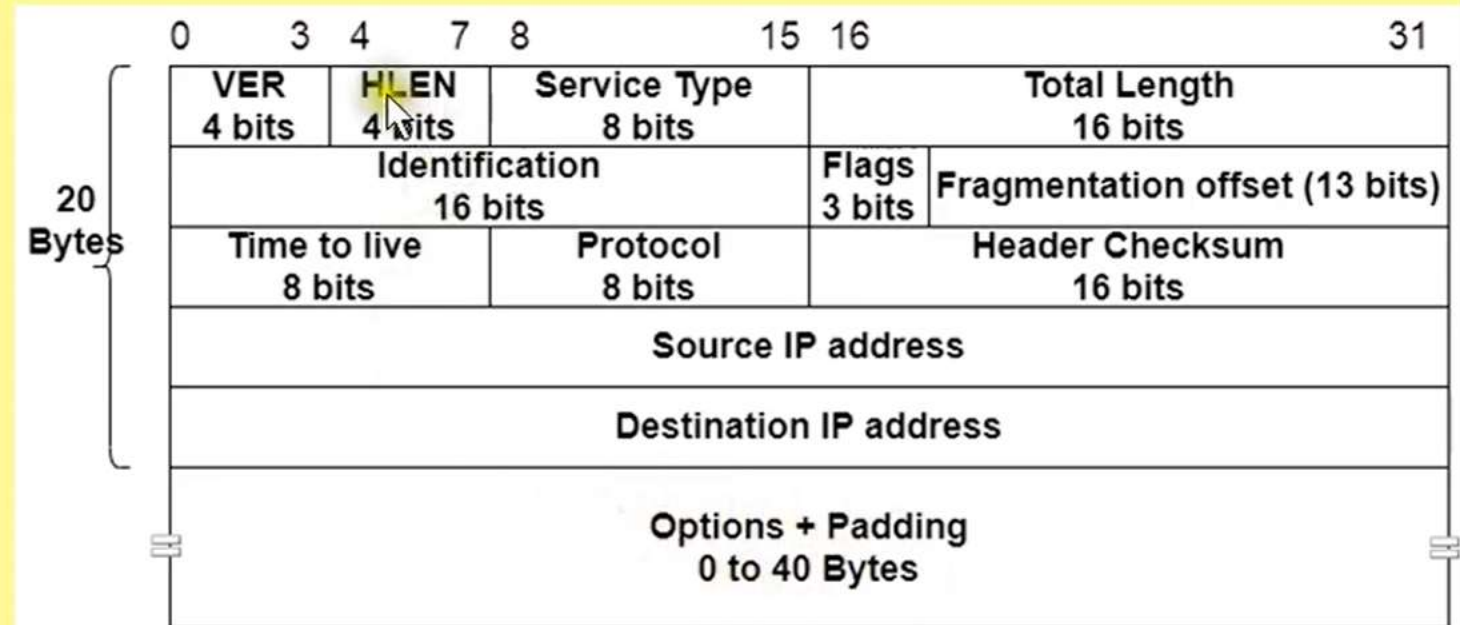
With options, the IP header

Size will be 20 bytes + 20 bytes = 40 bytes

Therefore,

HLEN = header size/ 4 bytes
= 40 bytes/ 4 bytes
= 10 D

HLEN in binary = 1010



IP Header Format

Answer:

HLEN = 10 D

HLEN = 1010

Q. The value of the total length field in an IP datagram is 36 and the value of the header length field is 5. How many bytes of data is the packet carrying ?

Solution:

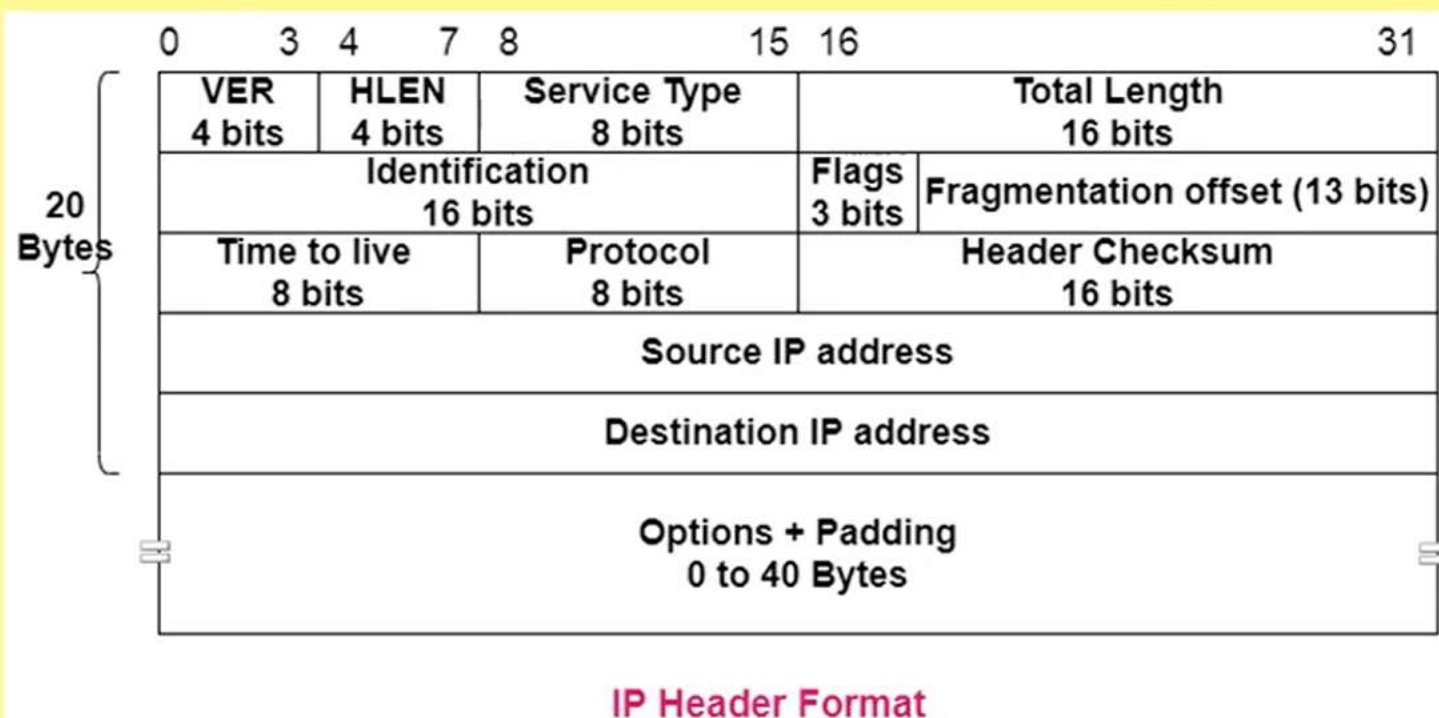
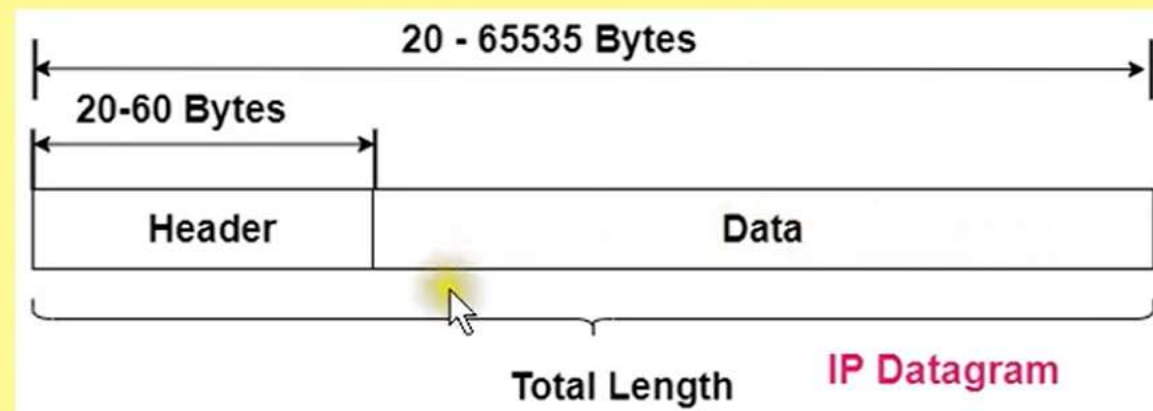
Total length = 36 bytes

HLEN = 5

Therefore,
header size = HLEN X 4 bytes
= 5 x 4 bytes
= 20 bytes

Therefore,
Length of data
= total length – header size
= 36 – 20
= 16 bytes

Answer:
Data = 16 bytes



Q. A datagram is carrying 1024 bytes of data. If there is no option information, what is the value of the header length field? What is the value of the total length field?

Solution:

Data length = 1024 bytes

Options = 0

Therefore, Header size = 20 bytes (fixed)

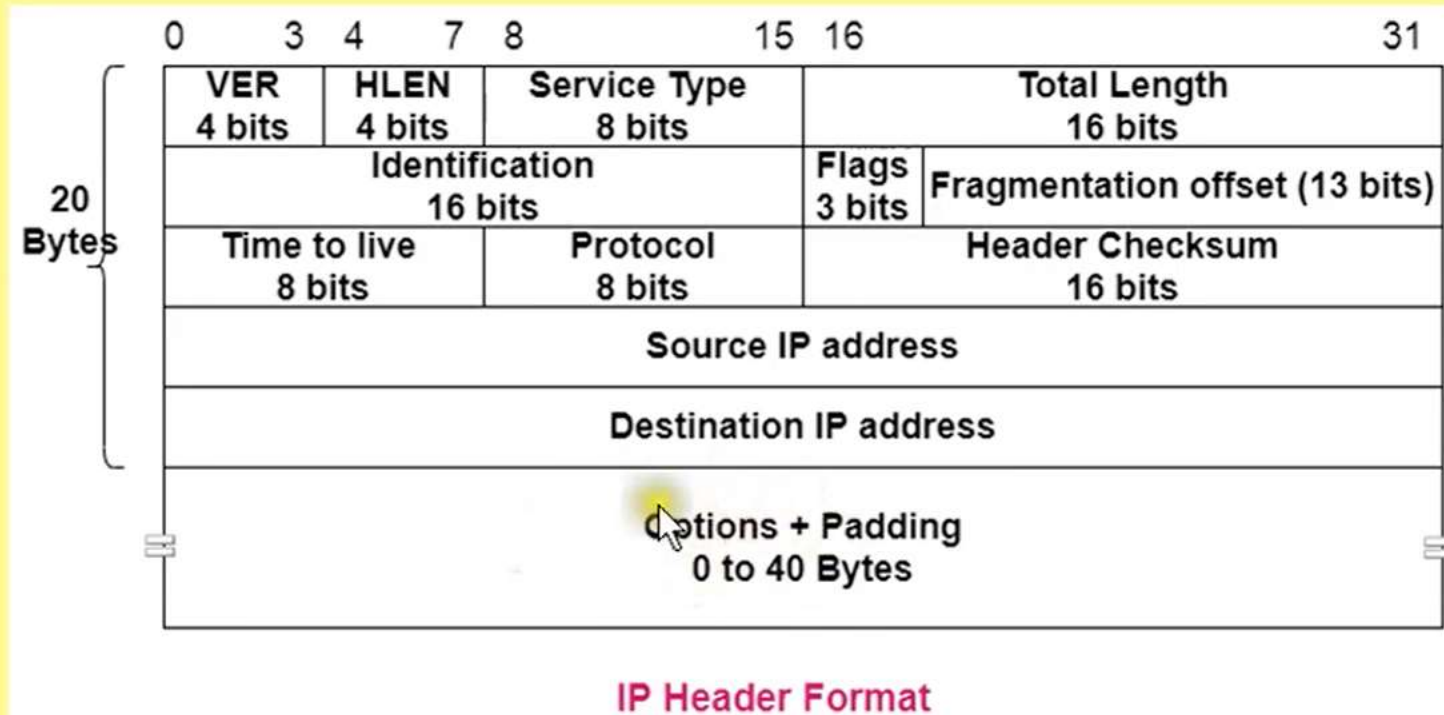
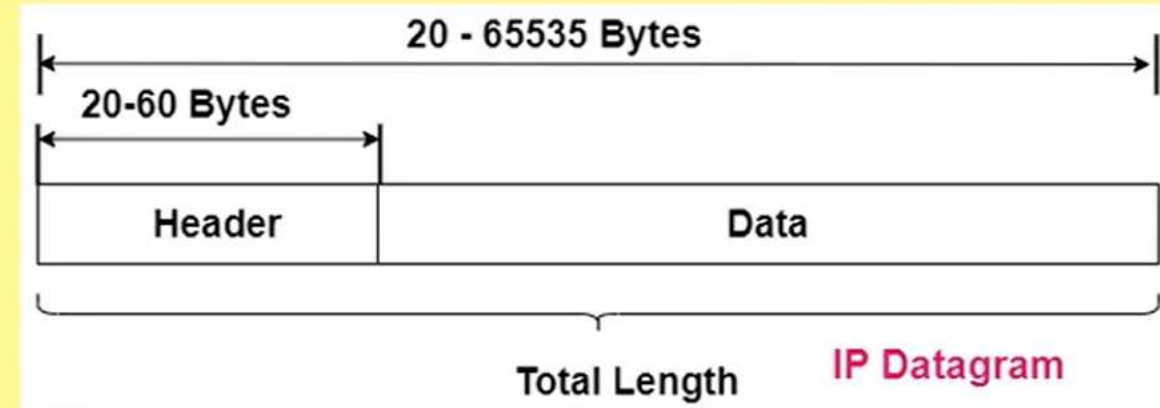
And HLEN = header size / 4bytes
= 20 bytes / 4 bytes
= 5 D

Total length = header size + Data
= 20 bytes + 1024 bytes
= 1044 bytes

Answer:

HLEN = 5 D = 0101 binary

Total length = 1044 bytes



Q. An IP datagram has arrived with the following information in the header (in hexadecimal)

45 00 00 54 00 03 00 00 20 06 00 00 7C4E 0302 B40E 0F02

Are there any options?

Is the packet corrupted ? Why?

Is the packet fragmented?

What is the size of the data?

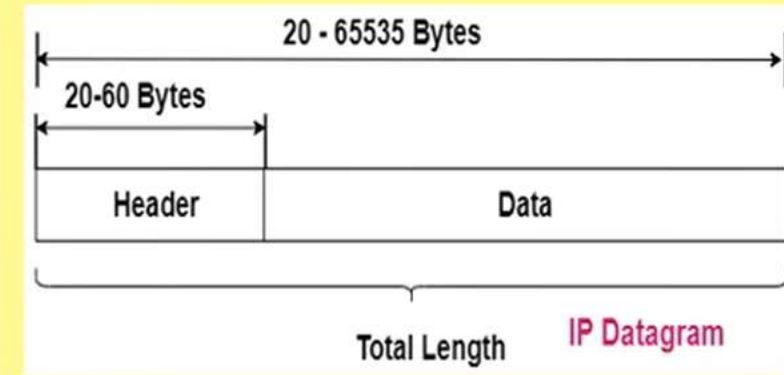
How many more routers can the packet travel to?

What is the identification number?

What is the type of service?

Unused	DF	MF
Flags		

Value	Protocol
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF



Solution:

4 5 00 0054 0003 0000 20 06 0000 7C4E0302 B40E0F02

VER = 4

HLEN = 5 Header size = HLEN x 4 bytes = 20 bytes --> options = 0 (No options)

ST = 00 → normal service

TL = 0054 = 84 D = 84 bytes, therefore data = TL – header size = 84 – 20 = **64 bytes**

IDENTIFICATION = **0003**

FLASGS/FRAG. OFFSET = 0000 → DF = 0 MF = 0 → **fragmentation not done**

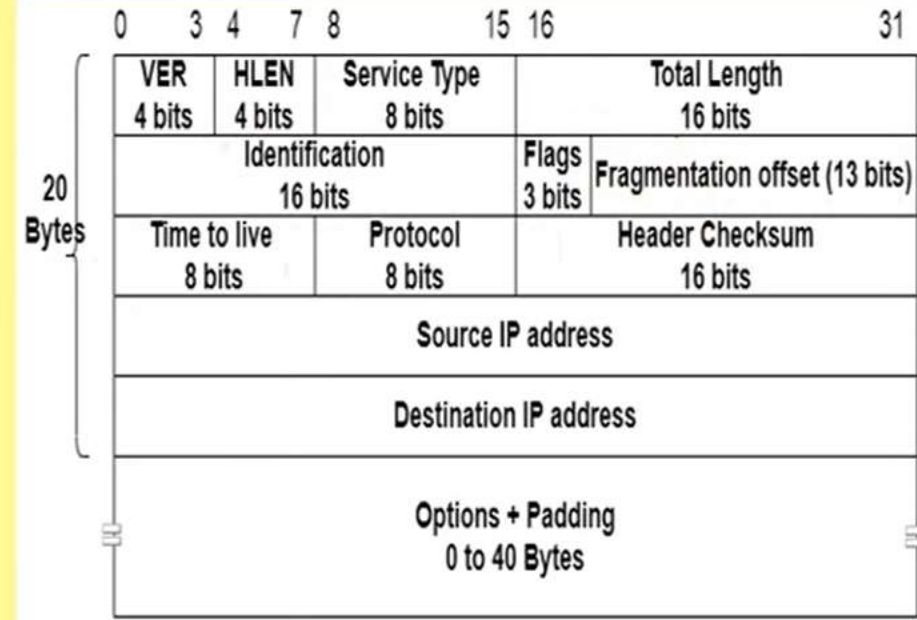
TTL = 20 → 32 D → packet can travel to **32 routers**

PROTOCOL = 06 → TCP data

CHECKSUM = 0000 → ?

SOURCE IP = 7C**4E**03**02** = 124.76.3.2

DESTINATION IP = B4**0E**0F**02** = 180.14.15.2



IP Header Format

IPv4 Checksum calculation:

4500 0054 0003 0000 2006 0000 7C4E0302 B40E0F02

Step1: arrange given hexadecimal dump of IPv4 header in 16 bit words

Step2: calculate sum of all 16 bit words excluding the checksum bits.

Step3: Find ones complement of the sum which will be the checksum.

4500
+0054
+0003
+0000
+2006
+7C4E
+0302
+B40E
+0F02

1A7BD

A7BD
+ 1

A7BE and ones complement of A7BE is 5841

5841 is the calculated checksum and the

checksum in the received IPv4 header is 0000.

This means that the packet is corrupted.

Figure 20.11 *Fragmentation example*

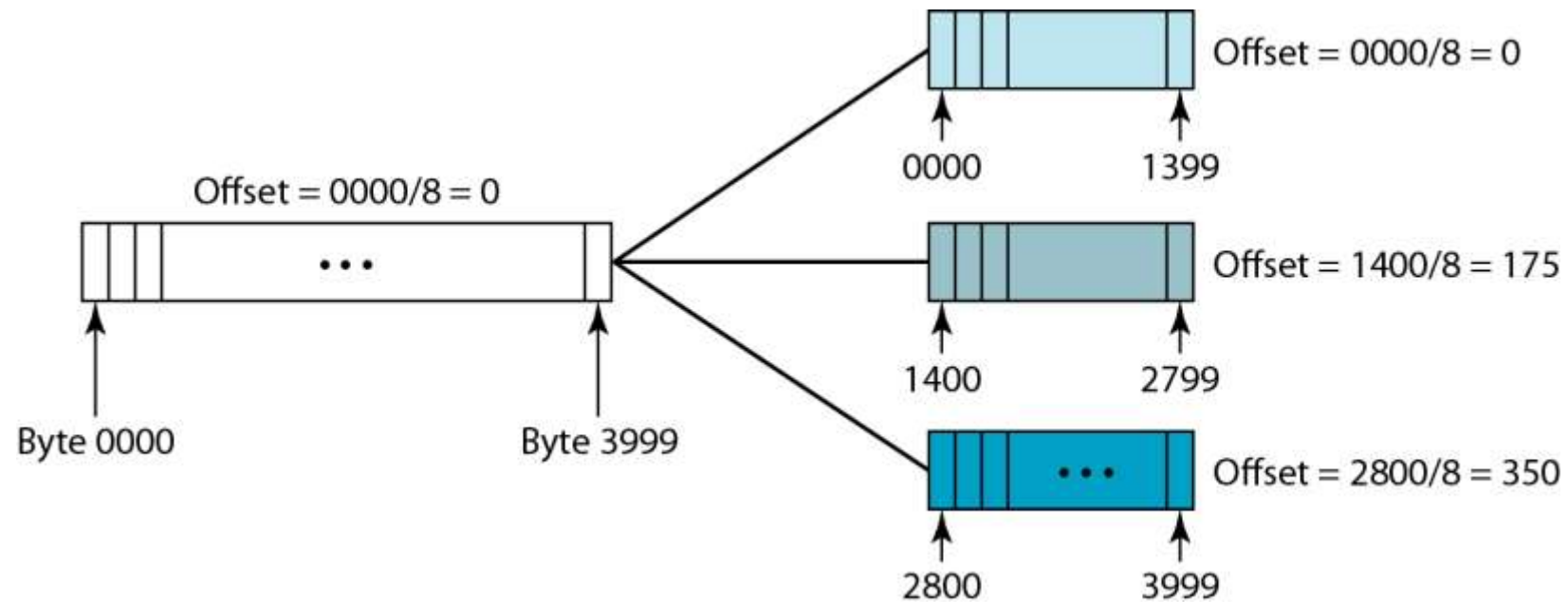


Figure 20.12 Detailed fragmentation example

